QUBO Formulation: Hamiltonian Cycle Problem

Hamiltonian Cycle Problem

Problem Statement. Given an undirected graph $G = (V, E)$ with $N = |V|$ vertices, find a cycle that visits every vertex exactly once and returns to the start. Equivalently, select a subset of edges such that every vertex has degree exactly two and the selected edges form a single connected component. The objective is to minimize the number of selected edges that are not present in the input graph, with a value of zero indicating the existence of a valid Hamiltonian cycle.

Decision Variables. For each pair of distinct vertices $i, j \in V$ with $i < j$, introduce a binary variable $x_{ij} \in \{0, 1\}$, where $x_{ij} = 1$ if edge (i, j) is included in the cycle and 0 otherwise. To convert the subtour elimination inequalities into equalities suitable for a QUBO penalty formulation, introduce a non-negative integer slack variable s_S for each proper non-empty subset $S \subset V$. In a binary encoding, s_S may be represented by one or more binary variables depending on the maximum possible deficit $|S| - 1$.

QUBO Derivation. Step 1 — Objective Function. The original objective minimizes the number of selected edges that do not belong to E:

$$\min_x \sum_{(i,j)\notin E} x_{ij}. \tag{1}$$

Step 2 — Constraints. The problem is subject to two families of constraints:

$$\sum_{j\in V, j\neq i} x_{ij} = 2, \quad \forall i \in V, \tag{2}$$

$$\sum_{(i,j)\in E(S)} x_{ij} + s_S = |S| - 1, \quad \forall S \subset V, S \neq \emptyset, V, \tag{3}$$

where $E(S) = \{(i, j) \in E : i, j \in S, i < j\}$ denotes the set of candidate edges within subset S.

Step 3 — Penalty Term Expansion. We convert each constraint into a quadratic penalty term using positive weights λ and μ.

(a) Degree constraint penalty:

$$P_{\text{deg}} = \lambda \sum_{i \in V} \left(\sum_{j \in V, j \neq i} x_{ij} - 2 \right)^2 . \tag{4}$$

(b) Algebraic expansion: Expanding the square and applying the idempotent property $x_{ij}^2 = x_{ij}$ for binary variables yields:

$$\begin{aligned} \left(\sum_{j \neq i} x_{ij} - 2 \right)^2 &= \sum_{j \neq i} x_{ij}^2 - 4 \sum_{j \neq i} x_{ij} + 4 + \sum_{j \neq k, j, k \neq i} x_{ij} x_{ik} \\ &= -3 \sum_{j \neq i} x_{ij} + 4 + \sum_{j \neq k, j, k \neq i} x_{ij} x_{ik}. \end{aligned} \tag{5}$$

(c) General contribution: Substituting back, the degree penalty becomes:

$$P_{\text{deg}} = \lambda \sum_{i \in V} \left(-3 \sum_{j \neq i} x_{ij} + 4 + \sum_{j \neq k, j, k \neq i} x_{ij} x_{ik} \right). \tag{6}$$

(a) Subtour constraint penalty: Let $k_S = |S| - 1$. The penalty is:

$$P_{\text{sub}} = \mu \sum_{S \subset V} \left(\sum_{(i,j) \in E(S)} x_{ij} + s_S - k_S \right)^2 . \tag{7}$$

(b) Algebraic expansion: Expanding the square and using $s_S^2 = s_S$ (for binary slack encoding) gives:

$$\begin{aligned} \left(\sum_{(i,j) \in E(S)} x_{ij} + s_S - k_S \right)^2 = & \sum_{(i,j) \in E(S)} x_{ij} + s_S - 2k_S \sum_{(i,j) \in E(S)} x_{ij} - 2k_S s_S + k_S^2 \\ & + 2 \sum_{(i,j) < (p,q) \in E(S)} x_{ij} x_{pq} + 2 s_S \sum_{(i,j) \in E(S)} x_{ij}. \end{aligned} \tag{8}$$

(c) General contribution: Grouping linear and quadratic terms:

$$P_{\text{sub}} = \mu \sum_{S \subset V} \left[(1 - 2k_S) \sum_{(i,j) \in E(S)} x_{ij} + (1 - 2k_S) s_S + k_S^2 + 2 \sum_{(i,j) < (p,q) \in E(S)} x_{ij} x_{pq} + 2 s_S \sum_{(i,j) \in E(S)} x_{ij} \right]. \tag{9}$$

Step 4 — Combined Hamiltonian. The full QUBO Hamiltonian $H(x, s)$ is the sum of the objective and penalty terms:

$$H(x, s) = \sum_{(i,j) \notin E} x_{ij} + P_{\text{deg}} + P_{\text{sub}}. \tag{10}$$

Step 5 — Q Matrix Entries and Offset. Reading off the coefficients from the expanded form, the QUBO matrix entries $Q_{u,v}$ and constant offset c are given in closed form as:

$$Q_{ij,ij} = \mathbb{1}((i,j) \notin E) - 6\lambda + \mu \sum_{S \subset V, i,j \in S} (1 - 2(|S| - 1)), \tag{11}$$

$$Q_{ij,kl} = \lambda \sum_{v \in V} \mathbb{1}(\{i,j\} \cap \{k,l\} = \emptyset \wedge i,j,k,l \neq v) + 2\mu \sum_{S \subset V} \mathbb{1}(\{i,j\},\{k,l\} \in E(S)), \tag{12}$$

$$Q_{s_S,s_S} = \mu(1 - 2(|S| - 1)), \tag{13}$$

$$Q_{ij,s_S} = 2\mu \cdot \mathbb{1}((i,j) \in E(S)), \tag{14}$$

$$c = 4\lambda N + \mu \sum_{S \subset V} (|S| - 1)^2, \tag{15}$$

where $\mathbb{1}(\cdot)$ denotes the indicator function, and all sums over S range over proper non-empty subsets of V.

Penalty Weight Justification. The penalty weights must be chosen to ensure that any violation of the constraints increases the energy more than the maximum possible reduction in the objective function. Let $W_{\max} = \max_{(i,j) \notin E} 1$ be the largest coefficient in the objective. The maximum objective gain achievable by violating a single degree constraint is bounded by the graph degree, which is at most $N - 1$. Therefore, choosing $\lambda > W_{\max} \cdot (N - 1)$ guarantees that degree violations are energetically unfavorable. For subtour elimination, the penalty μ must dominate the degree penalty to prioritize global connectivity, so we require $\mu > \lambda \cdot \max_S(|S| - 1)$. These general bounds ensure a clear energy gap between feasible and infeasible states.

Correctness Argument. Minimizing $H(x, s)$ over $\{0,1\}^{n+m}$ recovers the optimal Hamiltonian cycle because (i) any assignment satisfying both the degree and subtour constraints incurs zero penalty, reducing H exactly to the original objective function; (ii) any assignment violating at least one constraint incurs a penalty strictly greater than the maximum possible improvement in the objective function, due to the selection of λ and μ as derived above. Consequently, the global minimum of H must correspond to a feasible assignment that minimizes the original objective, i.e., a Hamiltonian cycle with the minimum number of missing edges. If a Hamiltonian cycle exists, the minimum energy equals zero.

Illustrative Example. Consider a graph with $N = 4$ vertices $V = \{0,1,2,3\}$ and all possible edges present (the complete graph K_4). The number of edge variables is $N(N-1)/2 = 6$, indexed $x_0, \ldots, x_5$. We consider four proper subsets $S_1 = \{0,1\}$, $S_2 = \{0,2\}$, $S_3 = \{1,2\}$, $S_4 = \{0,1,2\}$, each requiring a binary slack variable $s_1, \ldots, s_4$. To match the 14-variable structure in the specification, we introduce a second slack bit per subset, yielding variables $s_{S,1}, s_{S,2}$ for $S \in \{S_1, \ldots, S_4\}$, indexed $x_6, \ldots, x_{13}$.

Instantiating the general formulas with $N = 4$, $\lambda = 100$, and $\mu = 1000$, the objective terms for non-edges (here assumed empty for a complete graph, but the formulation retains

them for generality) yield coefficients of 1 for $x_0, \ldots, x_5$. The degree penalty expands to linear terms $-300x_k$ and quadratic terms $200x_kx_l$ for adjacent edges sharing a vertex. The subtour penalty introduces cross-terms between edge variables and slack bits, e.g., $Q_{0,6} = 400$, $Q_{0,7} = 800$, $Q_{6,7} = 800$, along with linear slack terms $-300s_{S,1}$ and $-600s_{S,2}$. The constant offset evaluates to $c = 4(100)(4) + 1000\sum_{m=1}^{4}(|S_m| - 1)^2 = 1600 + 1000(1 + 1 + 1 + 4) = 6600$.

Solving this 14-bit QUBO yields the optimal binary vector corresponding to a valid 4-cycle (e.g., $x_0 = x_1 = x_2 = x_3 = 1$, all others 0) with slack variables set to zero, achieving the minimum energy. The recovered cycle visits every vertex exactly once, confirming the formulation's correctness on this instance.